

First steps in the Linux terminal

Laboratory 01

Work in Bash on the laboratory Linux workstation. The tasks form one sequence: keep your files for the next task. Use the terminal for file operations and use the lecture and installed documentation when needed.

Before you start: the workstation and Taurus

The **terminal** is the text interface; the **shell** interprets your commands. A local shell runs on the workstation. **SSH** opens a shell on another computer: after connecting to Taurus, commands in that session run on Taurus. Use **hostname** to identify the host and **whoami** to identify the account.

Your laboratory account's **home-directory files are shared** between the Linux workstations and Taurus. A saved change made through one host is therefore visible through the other. SSH does not copy or synchronise these files; the shared storage is a feature of the university setup. The hosts can have different software and separate running programs. Your home computer does not automatically share this storage: SSH gives remote access, while downloading creates a separate copy.

Connecting: use `ssh YOUR_LOGIN@taurus.fis.agh.edu.pl`, replacing `YOUR_LOGIN` with your assigned username. If asked to confirm the server identity, check the displayed fingerprint with the instructor before accepting. Password input normally shows no characters. **exit** ends the remote shell and returns you to the local shell; it does not shut down Taurus.

Account and access information: WFiS Taurus instructions.

Task 1.01 Two hosts, one home directory

1. Open **local terminal A**. Run **hostname** and **whoami**; note the results. Go home, create a *new* directory `unix-lab01`, and enter it. Using **echo** and **>**, create `connection.txt` containing **Created on the workstation**. Display the file. If the directory already exists, choose a fresh name and use it throughout this worksheet.
2. Open **terminal B** on the workstation and connect to Taurus using SSH. Repeat **hostname** and **whoami**. Which result identifies the change of host? Display your current directory with **pwd**.
3. In **remote B**, enter `unix-lab01` inside your home directory. Read `connection.txt`, then append **Checked on Taurus** using **echo** and **>**. In **local A**, read the file again. Explain why both lines are visible even though you have not copied the file.
4. In B, use **exit**, then **hostname** to verify your return. Continue the remaining tasks in **local A**. Explain the difference between the host running a command and the storage containing the file it reads.

Navigation and documentation

Task 1.02 Build and navigate your workspace

1. Display your current directory. Go to `/tmp`, then to its parent; display your location after each move. Return home. Is `/root` the same directory as `/`? Display `$HOME` once as its value and once as literal text.
2. Enter the `unix-lab01` directory created in Task 1.01. Create `data`, `results`, `notes` and `scratch` inside it. Also create `simulations/run01/output`, including its missing parents.
3. Create an empty file `notes/session.txt`. Display your workspace with `tree`, limited to two levels and then to three. What becomes visible at the third level?
4. Enter `data`, then reach `results` using a relative path. Return to the previous directory with `cd -`, then to `unix-lab01` using `...` Use Tab completion at least once.

If `tree` is unavailable, inspect the named directories with `ls`; no installation is required.

Task 1.03 Use the manual to choose and verify options

`man ls` opens the installed reference for `ls`. With the usual `less` viewer: type `/word` and press Enter to search, `n` for the next match, `Space/b` for the next/previous screen, and `q` to return to the shell. Read the surrounding description, not just the matching word.

1. Open `man ls` and find **SYNOPSIS**. Explain the square brackets and the ellipsis in `ls [OPTION] ... [FILE] ...`. Must a filename be supplied?
2. Use the manual to find options for including all entries (even `.` and `..`) and for listing a directory itself instead of its contents. Starting from `ls .`, predict and test each change separately. State the option, its purpose, and an observation that confirms its effect. Why do `.` and `..` appear when you include all entries?
3. Find the option for a long listing. Display all entries in your home directory, including hidden ones, and a long listing of `/etc`. Identify a name, size, owner and file type; you do not need to decode permissions yet.
4. Compare the **NAME** and description sections of `man 1 passwd` and `man 5 passwd`. Which page describes a command and which a file format? Explain why a manual reference needs both a name and a section number. Reading these pages does not change your password.
5. Use `help cd` to check what `cd` with no argument does. Why is Bash's built-in help appropriate for this command?

Use documentation again later: in Task 1.05, find the required copying options in `man cp` before testing them. If a manual page is unavailable, ask the instructor; `ls -help` and `cp -help` provide option summaries.

A small measurement project

At the start of each task below, return to `~/unix-lab01`. Relative paths are interpreted from your current directory; a step that changes directory also changes that starting point.

Task 1.04 Create and inspect your data

1. Use `echo`, `>` and `»` to create `data/motion.txt` with exactly the following contents. Recall and edit earlier commands with the arrow keys.

```
# time_s position_m
0.0 0.00
0.1 0.03
0.2 0.12
0.3 0.27
0.4 0.48
```

2. Display the file with `cat` and `tac`. Where does the header appear in each output? Does either command change the file?
3. Display its first three lines, its final two lines, and then all data rows without the header. Explain why `tail -n 2` and `tail -n +2` differ.
4. Open the file with `less`; search for `0.12`, go to the end, return to the beginning, and quit. Which navigation keys also worked in `man`?

Task 1.05 Copy, rename and recover data

1. Copy `data/motion.txt` into `results`. Rename the copy to `trial.txt`, then move it into `scratch`. After each operation, check where the file exists. Keep the original in `data` unchanged.
2. Copy the entire `data` directory to a new directory `data-backup`. Find an appropriate option in `man cp`. Predict what repeating the same copy command would do now that the destination exists; justify your answer using the manual, without repeating the command.
3. Overwrite only `scratch/trial.txt` with the line `BAD RUN`. Confirm that the original and backup still contain the measurements.
4. Restore `scratch/trial.txt` from the backup. Use an option that asks before overwriting an existing file: find it in `man cp`, then confirm the replacement. Inspect the restored contents.

Predict, diagnose, recover

Task 1.06 File contents, timestamps and redirection

Work on `notes/test.txt`; leave the measurement file unchanged.

1. Create `notes/test.txt` containing two lines: `alpha` and `beta`. Predict the effect of `touch` on this existing file. Run `touch notes/test.txt` and inspect the contents. What is `touch` intended to update?
2. Append `gamma`, inspect the file, then write `delta` using `>` and inspect it again. Explain the difference.
3. A colleague proposes `cat notes/test.txt > notes/test.txt` to “save” the file. Explain the problem *before* testing it on this disposable file. Inspect the result. Why can `cat` no longer read the previous text?
4. Re-create the file with one line, `recovered`. Run `tail -f notes/test.txt`. Stop it with Ctrl+C. Then run `clear` and check that the file still exists and still contains that line.

Task 1.07 Hidden entries, pathnames and deletion

1. Create `scratch/cleanup`, then create an empty file `scratch/cleanup/.keep`. Compare ordinary `ls` with a listing that includes hidden entries.
2. Try to remove `scratch/cleanup` with `rmdir`. Explain the error. Remove its file using `rm -i`, then remove the now-empty directory.
3. Create `notes/run summary.txt` containing `Run complete`. Display it. Why does the space need quoting or escaping?
4. From the workspace, deliberately try `cd Data`. Explain the error, then enter the actual data directory and return to the workspace.

Checkpoint: a hidden file is still a file. Does the leading dot make it private?

Complete and verify your project

Task 1.08 Prepare and verify a results package

Return to your workspace. Complete this task without copying a full solution from the lecture; documentation is allowed.

1. Create **results/final**. From the original measurement file, save the header to **header.txt**, all five data rows to **values.txt**, and the last measurement row to **last.txt**, all inside this new directory.
2. Combine **header.txt** and **values.txt**, in that order, into **reconstructed.txt** in the same directory, using one **cat** command.
3. Create **README.txt** there with two lines identifying the input path and the units. Display a detailed listing with human-readable sizes and inspect the generated text files.
4. Show the instructor that the original is unchanged, the reconstructed file has the same six lines, and **last.txt** contains `0.4 0.48`. Explain how you checked these properties. Before leaving, explain what **exit** would end in your current terminal; use it only when you have finished any optional work.

Optional extension: the filesystem hierarchy

Use read-only commands outside your workspace. Work in pairs after completing Task 1.08: inspect first, then justify your answers using the lecture and what you see.

1. Run **ls -ld /bin /sbin /lib**. What does your system show? If arrows appear, where do they lead? You do not need to create any links.
2. Inspect **/etc**, **/usr** and **/var** with **ls**. For each location, name one entry you can see and relate it to the directory's purpose.
3. Run **ls -l /dev/null**, **head -n 5 /proc/meminfo**, and **ls /sys/class/net**. Which output describes a special file, memory, and network interfaces? Do not attempt to modify these locations.
4. Discuss: where would you keep your course work, system configuration, installed software, and changing service logs? Why is **/tmp** a poor place for the only copy of your results? Does one directory tree imply one physical disk?

Additional challenge: a name beginning with a hyphen. In **scratch**, create a file named **-sample.txt**. Display its detailed entry without treating its name as an option. Remove only this file, then return to the workspace.